

How are oceans formed?

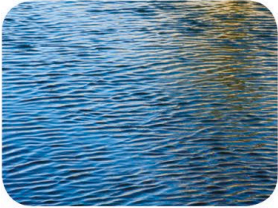
Our planet is a wonderfully watery world. Oceans cover more than 70 percent of the Earth's surface, and contain 97 percent of the total water supply on Earth. The oceans formed almost 4 billion years ago, when low-lying areas became filled with rainwater.

Moving plates

Earth's surface is made up of a thin crust, which covers the top of the hot, rocky mantle. Together, the crust and mantle make moving pieces of land, called tectonic plates. New ocean floor develops where these plates pull apart.

Making waves

Ocean waves are influenced by wind speed, distance, and direction. Their size and shape range from tiny ripples to giant swells.



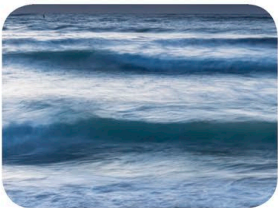
Ripple

A gentle wind blowing over the ocean results in small ripples.



Chop

Stronger winds make rough waves known as chops.



Swell

Large waves, known as swells, create a more regular wave pattern over time.

Tectonic plates slowly drift apart over time.

Heat from the Earth's core sends rock up to the surface.

